

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING IV B.TECH II SEM TITLE
and ABSTRACT SUBMISSION FORM Major Project-Mern Stack**

Title of the Application	Vision Grow-Smart Vision for Plant Health
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Guide Name: Dr. Naga Chandrika Vaidya**Abstract:**

Plant diseases pose a serious threat to agricultural productivity and food security, making early detection and accurate assessment essential for effective crop management. This project proposes an automated plant disease detection system using image processing and machine learning techniques to identify diseases from leaf images. The system preprocesses input images to enhance quality and extracts important features such as color, texture, and shape using a convolutional neural network (CNN). Based on these features, the model classifies the type of plant disease and distinguishes between healthy and infected regions of the leaf. In addition to disease identification, the system calculates the percentage of disease infection by segmenting the affected area from the total leaf area. This helps in determining the severity level of the disease. Based on the detected disease type and infection percentage, suitable preventive and control techniques are suggested, including proper pesticide application, organic remedies, and improved farming practices. The proposed system supports timely decision-making, minimizes crop loss, and promotes sustainable agriculture through precise and efficient disease management.

Signature of the Guide